# 15 Tech Breakthroughs of 2021–2026

Research report · as of 2026-05-08 · AI · Bio · Energy · Space · Quantum

This report tracks 15 technology breakthroughs that broke through between 2021 and 2026 — from transformers and reasoning models to GLP-1 agonists, CRISPR therapies, sub-threshold logical qubits, the Mechazilla booster catch, the collapse of battery costs below $110/kWh, and commercial driverless robotaxis. For each breakthrough we identify the people behind it, the component *sub-ideas* that make it up, the cutting edge of the last 12–18 months, the rising organizations and individuals to watch, and what to look for next. All claims are footnoted to primary sources where possible. Where a claim could not be independently re-verified, we have flagged it.

## Latest weekly highlights

**Week ending 2026-05-08.** Intra-day supplement to the morning's run: three primary-source items the AM scan missed. Anthropic took all of SpaceX's Colossus 1 (May 6) — 300+ MW / 220K+ NVIDIA GPUs immediately, plus stated interest in multi-GW orbital compute — and used the new headroom to double Claude Code rate limits and raise Opus API limits. Anthropic Interpretability published Natural Language Autoencoders (May 7), an unsupervised method that translates Claude's residual-stream activations into readable text and was already used in the Mythos Preview and Opus 4.6 alignment audits. OpenAI opened GPT-5.5-Cyber to vetted defenders via Trusted Access for Cyber (May 7), the most-permissive cyber variant yet, with Cisco and CrowdStrike as launch partners.

• **Anthropic + SpaceX Colossus 1 deal: 300+ MW / 220K+ GPUs in a month** — Anthropic signed for the entire compute capacity of SpaceX's Colossus 1 data center in Memphis — 300+ MW (over 220,000 NVIDIA GPUs) coming online within the month — and immediately doubled Claude Code five-hour rate limits and raised Opus API limits; the agreement also includes stated interest in multi-gigawatt orbital AI compute with SpaceX.

• **Anthropic Natural Language Autoencoders read Claude's activations as text** — Anthropic Interpretability released Natural Language Autoencoders (NLAs) — verbalizer/reconstructor LLM pair, RL-trained for round-trip activation→text→activation fidelity — that surface unverbalized evaluation-awareness on 16-26% of safety/SWE-bench transcripts (vs <1% on real opt-in usage) and lift hidden-motivation auditing from <3% to 12-15%; deployed in Mythos Preview and Opus 4.6 pre-deployment audits, with code and Neuronpedia frontend released.

• **OpenAI GPT-5.5-Cyber via Trusted Access for Cyber program** — OpenAI rolled out GPT-5.5-Cyber, the most-permissive variant of GPT-5.5 for vetted security defenders (bug-hunting, malware analysis, reverse engineering allowed; credential-theft and malware-writing still blocked), gated via Trusted Access for Cyber identity tier with Cisco and CrowdStrike among the launch partners; Axios sources put offensive-cyber capability roughly on par with Anthropic Mythos Preview.

## Executive summary

Breakthroughs at a glance: top named figures (org-type tagged), current state, and one rising organization to watch.

| # | Breakthrough | Domain | Top people | Current state (1 line) | Watch this |
|---|---|---|---|---|---|
| **01** | **Transformers / LLMs / coding agents** | AI | Ashish Vaswani (Biz)<br>Noam Shazeer (Biz) | Mixture of Experts (MoE) architecture has become the standard for frontier models, with DeepSeek-V3 activating only 5.5% of parameters per token, enabling efficient trillion-par… | **Cursor (Anysphere)** |
| **02** | **RLHF + reasoning models (test-time compute scaling)** | AI | Richard Sutton (Cross)<br>Andrew Barto (Acad.) | DeepSeek-R1 demonstrates that pure reinforcement learning without supervised fine-tuning can achieve o1-level reasoning performance, published under MIT license with visible cha… | **xAI Grok 5** |

| # | Breakthrough | Domain | Top people | Current state (1 line) | Watch this |
|---|---|---|---|---|---|
| 03 | **Diffusion models (image + video generation)** | AI | David Holz (Biz)<br>Emad Mostaque (Biz) | Runway Gen-4.5 (December 2025) demonstrates advanced physics understanding including momentum, weight, fluid dynamics, and object interactions, with native audio generation inte… | **Stability AI Cascade Model** |
| 04 | **AlphaFold + computational protein design** | BIO | Demis Hassabis (Biz)<br>John Jumper (Biz) | AlphaFold 3 (published May 2024 in Nature) predicts structures of proteins, nucleic acids, ligands, ions and modified residues with substantially improved accuracy for protein-l… | **Chai Discovery** |
| 05 | **mRNA vaccines + LNP delivery** | BIO | Katalin Karikó (Acad.)<br>Drew Weissman (Acad.) | Moderna and Merck's mRNA-4157 personalized cancer vaccine combined with Keytruda sustained 49% reduction in recurrence/death risk in melanoma at 5-year follow-up (announced Janu… | **Moderna/Merck cancer vaccine program** |
| 06 | **GLP-1 agonists** | BIO | Jens Holst (Acad.)<br>Mads Krogsgaard Thomsen (Biz) | Retatrutide (GIP/GLP-1/glucagon triple agonist) showed 28.7% weight loss and A1C reductions up to 2.0% in Phase 3 TRANSCEND-T2D-1 trial (announced March 2026). | **Retatrutide clinical program (Eli Lilly)** |
| 07 | **CRISPR therapies (Casgevy + base/prime editing)** | BIO | Feng Zhang (Acad.)<br>David Liu (Acad.) | Casgevy (exagamglogene autotemcel) FDA-approved December 8, 2023, for sickle cell disease in patients ≥12 years; uses CRISPR/Cas9 to induce fetal hemoglobin, eliminating vaso-oc… | **Prime Medicine (prime editing platform)** |
| 08 | **Quantum error correction (logical qubits below threshold)** | QUANTUM | Hartmut Neven (Biz)<br>John Preskill (Acad.) | Google Willow (December 2024): 105-qubit processor achieved surface code error suppression below threshold with distance-7 code at 0.143% per-cycle logical error rate; first dem… | **PsiQuantum** |
| 09 | **Reusable rockets (Falcon 9 + Starship catch)** | SPACE | Elon Musk (Biz)<br>Gwynne Shotwell (Biz) | Falcon 9 Booster Reusability Record (May 2026): 384+ booster reflights with 100% success rate; single booster B1062 reached 22 flights by early 2026; turnaround as short as 3 we… | **Blue Origin New Glenn** |
| 10 | **Renewables + battery cost collapse** | ENERGY | Jay Whitacre (Cross)<br>Tejas Patel (Biz) | BloombergNEF Battery Price Collapse (June 2025): Lithium-ion pack prices fell 8% to record low of $108/kWh; stationary storage at $70/kWh (45% drop YoY); BEV packs at $99/kWh (2… | **BYD Sodium-Ion** |
| 11 | **Self-driving cars / robotaxis** | AI | Tekedra Mawakana (Biz)<br>Dmitri Dolgov (Biz) | Waymo One scaled paid driverless rides from a single Phoenix geofence (2020) to commercial operation across Phoenix, San Francisco, Los Angeles, Austin, and Atlanta, surpassing … | **Wayve** |
| 12 | **Humanoid robotics** | ROBOTICS | Sergey Levine (Biz)<br>Jitendra Malik (Acad.) | Figure's Helix 02 (Jan 2026) shows unified full-body autonomy from a single VLA — dishwasher loading, bottle uncapping, pill extraction, syringe dispensing — without task-specif… | **Figure AI** |
| 13 | **Fusion energy** | ENERGY | Omar Hurricane (Gov)<br>Bob Mumgaard (Biz) | NIF achieved 10+ ignition shots with a peak yield of 8.6 MJ (April 2025) — establishing repeatable net energy gain and a target-gain trend above 4:1 for inertial confinement fus… | **Commonwealth Fusion Systems (SPARC)** |

| # | Breakthrough | Domain | Top people | Current state (1 line) | Watch this |
|---|---|---|---|---|---|
| **14** | **Brain-computer interfaces** | BIO | Leigh Hochberg (Cross)<br>Edward Chang (Acad.) | Neuralink's PRIME trial has implanted 5+ patients across multiple sites by mid-2025; the second patient ('Alex') achieved CAD control and gaming within days, and Blindsight (vis… | **Neuralink**<br>**Blindsight** |
| **15** | **AI accelerators / advanced compute** | HARD WARE | Jensen Huang (Biz)<br>Andrew Feldman (Biz) | NVIDIA Rubin GPUs ship in 2H 2026 with ~3× Blackwell performance and 1.2 EFLOPS FP8 training capability; Rubin Ultra follows in 2027. | **Tenstorrent** |

# Transformers / LLMs / coding agents

**Impact so far.** Transformers and LLMs have enabled agentic AI systems that autonomously solve software engineering tasks, with frontier models now matching or exceeding professional developer capabilities on real-world codebase tasks.

**Watch for next.** Watch for scaling reasoning with test-time compute to surpass human performance on complex multi-step planning; emergence of true agentic systems that coordinate multiple specialized coding agents in parallel; and development of interpretable reasoning traces that enable verification and debugging of AI agent decisions.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Ashish Vaswani**[1] | Co-founder and CEO, Essential AI; Co-author of 'Attention Is All You Need'<br>Essential AI | Biz | — |
| **Noam Shazeer**[2] | VP Engineering, Gemini Co-lead<br>Google DeepMind | Biz | — |
| **Dario Amodei**[3] | CEO<br>Anthropic | Biz | — |
| **Ilya Sutskever**[4] | CEO<br>Safe Superintelligence Inc. (SSI) | Biz | — |
| **Jeff Dean**[5] | Chief Scientist, Co-lead Gemini<br>Google DeepMind | Biz | — |
| **Yann LeCun**[6] | Founder and Chief Scientist<br>Advanced Machine Intelligence Labs (AMI Labs); Jacob T. Schwartz Chaired Professor, NYU Courant Institute | Cross | ★ ACM A.M. Turing Award 2018 |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Coding agents**[7] | Agentic LLMs that autonomously edit, run, and debug code have shifted programming from write-then-run to delegate-then-review — Cursor, Devin, Claude Code, and GitHub Copilot Workspace are now standard tooling for many engineering teams. | Cursor · Devin (Cognition) · Claude Code · GitHub Copilot Workspace · Aider |
| **Mixture-of-Experts (MoE)**[8] | Conditional computation that activates only a small fraction of parameters per token (DeepSeek-V3 ~5.5%) lets trillion-parameter models stay tractable to serve, decoupling parameter count from per-token cost. | DeepSeek-V3 · Mixtral 8x22B · Qwen3-MoE · Grok-1 |
| **Long-context windows**[9] | Production context lengths exploded from 4K (GPT-3.5) to 1M+ tokens (Gemini, Claude Sonnet 4, Qwen2.5-1M), enabling whole-codebase reasoning, long-document analysis, and persistent agent memory. | Gemini 2.5 Pro 2M · Claude Sonnet 4 1M · GPT-4.1 1M · Qwen2.5-1M |
| **Multimodal frontier models**[10] | Native multimodal training (vision + audio + text + code in one model) replaced bolt-on adapters and is now the default for frontier systems, supporting voice agents, screen-reading, and embodied use cases. | GPT-4o · Gemini 1.5/2.5 · Claude 3.5/4 Vision · Llama 3.2 Vision |
| **Open-weight catch-up**[11] | Open-weight models (Llama, Qwen, DeepSeek) closed most of the gap with closed-frontier systems, with DeepSeek-V3 and R1 published with permissive MIT-style licenses, restructuring the geopolitics of AI compute. | DeepSeek-V3 / R1 · Llama 3.x · Qwen2.5 / Qwen3 · Mistral Large |
| **Model Context Protocol (MCP)**[12] | Anthropic's open MCP standard (Nov 2024) became the lingua franca for connecting LLMs to external tools and data — adopted across Claude, OpenAI, and most agent frameworks, replacing one-off plugin systems with a single client/server protocol. | Anthropic MCP · MCP servers (Slack, GitHub, Linear, Notion) · Claude Desktop · OpenAI MCP support |

| Sub-idea | Description | Examples |
|---|---|---|
| **AWS MCP Server (managed)**[13] | Managed MCP server providing secure, auditable agent access to AWS services with file uploads, long-running operations, sandboxed Python, IAM guardrails, and CloudTrail logging. | Agents can call any AWS API via single tool · Sandboxed Python execution for multi-step operations · 40+ pre-built skills for IaC, storage, analytics, serverless |
| **GPT-Realtime-2 voice stack**[14] | Native real-time voice agent stack with GPT-5-class reasoning, 70+ language live translation, and streaming Whisper transcription, exposed via the OpenAI API. | Real-time voice agent with reasoning + tool use · Live multilingual translation in 70+ languages · Streaming low-latency transcription |
| **OpenAI GPT-5.5-Cyber (Trusted Access for Cyber)**[15] | May 7, 2026 — Domain-specialized variant of GPT-5.5 for vetted cyber defenders, launched via Trusted Access for Cyber identity tier. Permits bug-hunting, malware analysis, reverse engineering; blocks credential theft and malware writing. Cisco and CrowdStrike are launch partners; Axios sources peg vuln-finding capability ~ Anthropic Mythos Preview. | Vetted SOC and red-team teams at Cisco / CrowdStrike-tier orgs gain access to the more-permissive variant · Phishing-resistant auth required from June 1, 2026 for the highest access tier |

## b) Current research frontier (last 12–18 months)

**01.** Mixture of Experts (MoE) architecture has become the standard for frontier models, with DeepSeek-V3 activating only 5.5% of parameters per token, enabling efficient trillion-parameter deployment.[16]

**02.** Context length scaling to 1M tokens is now production-ready with Gemini 2.5 Pro (2M tokens), Claude Sonnet 4 (1M tokens), and open-source Qwen2.5-1M.[17]

**03.** Claude Opus 4.7 and GPT-5.5 represent 2026 frontier capabilities with improved instruction persistence and multi-step agentic orchestration.[18]

**04.** Claude Mythos Preview achieves 93.9% on SWE-bench Verified, though only 45.9% on the harder contamination-free SWE-bench Pro, revealing data contamination in benchmark evaluation.[19]

**05.** Diffusion transformers with patch-based representations have emerged as scalable architecture for both language and vision domains, with Next-Latent Prediction extending self-supervised learning to latent space.[20]

**06.** GPT-5.5 Instant produces 52.5% fewer hallucinated claims than GPT-5.3 Instant on high-stakes prompts in medicine, law, and finance (OpenAI release notes, May 5 2026).[21]

**07.** May 6, 2026 — Anthropic secures all of SpaceX Colossus 1 (~300 MW / 220K+ GPUs within the month), doubles Claude Code rate limits, raises Opus API limits, and expresses interest in multi-GW orbital compute with SpaceX.[22]

## c) Who could be next

**Cursor (Anysphere)** [STARTUP][23] — Anysphere Inc. · Series D, $29.3B valuation (2026)

Fastest-growing AI coding IDE, reaching $29.3B valuation with $2B+ ARR by early 2026, supporting parallel multi-agent coding.

**Cognition Labs (Devin)** [STARTUP][24] — Cognition Labs · Series C+, $400M at $10.2B valuation; targeting $25B (2026)

Autonomous AI software engineer scaling ARR from $1M to $73M in 9 months; acquired Windsurf; targeting $25B valuation in April 2026.

**Safe Superintelligence Inc. (SSI)** [STARTUP][25] — SSI · Series B, $2B at $32B valuation (April 2025)

Sutskever's new lab focused on safety-first AGI path with $3B+ raised and $32B valuation; significant backing from Alphabet and Nvidia.

**Advanced Machine Intelligence Labs (AMI Labs)** [STARTUP][6] — AMI Labs · Series A, $1.03B at $3.5B (March 2026)

Yann LeCun's new venture focused on world models and open-source development; raised $1.03B at $3.5B pre-money valuation in March 2026.

**Anthropic enterprise-AI JV** [STARTUP][26] — Anthropic / Blackstone / Hellman & Friedman / Goldman Sachs · $1.5B initial capitalization; backed by Blackstone, H&F, Goldman Sachs

First services-led commercialization vehicle for a frontier-lab model; converts mid-market deployments from pure API consumption to verticalized AI services.

## Flags / unverified

■ SWE-bench Verified benchmark scores may be inflated due to data contamination; SWE-bench Pro (contamination-free) shows significantly lower performance

■ Meta's Llama benchmark results were reportedly fudged according to departing chief scientist Yann LeCun

■ No verified information found on Tom Brown's current 2026 affiliation beyond his core role at Anthropic

# RLHF + reasoning models (test-time compute scaling)

**Impact so far.** Reasoning models with test-time compute scaling have fundamentally shifted AI capability distribution from training time to inference time, enabling smaller models to solve previously intractable problems in mathematics, coding, and science while achieving cost-competitive inference.

**Watch for next.** Monitor scaling laws for reasoning beyond current limits; emergence of multi-step verifiable reasoning that enables formal proof verification; and development of interpretable chain-of-thought that reveals genuine reasoning versus pattern matching in reasoning model outputs.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
| --- | --- | --- | --- |
| **Richard Sutton**[27] | Professor, University of Alberta; Senior Scientist, DeepMind<br>University of Alberta / DeepMind | Cross | ★ ACM A.M. Turing Award 2024 (announced 2025) |
| **Andrew Barto**[27] | Emeritus Professor<br>University of Massachusetts Amherst | Acad. | ★ ACM A.M. Turing Award 2024 (announced 2025) |
| **Ilya Sutskever**[25] | CEO<br>Safe Superintelligence Inc. | Biz | — |
| **Dario Amodei**[3] | CEO<br>Anthropic | Biz | — |
| **Demis Hassabis**[28] | CEO<br>Google DeepMind | Biz | — |
| **DeepSeek Research Team**[11] | Lead researchers on DeepSeek-R1<br>DeepSeek-AI | Biz | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
| --- | --- | --- |
| **Test-time compute scaling**[29] | Pioneered by OpenAI o1/o3, scaling inference-time tokens (often 10–100x more thinking) yields steeper accuracy gains than equivalent training-time scaling on math, code, and science benchmarks. | OpenAI o1 / o3 · Gemini 2.5 Thinking · Claude reasoning |
| **Pure-RL reasoning (R1-Zero recipe)**[11] | DeepSeek-R1 showed that reasoning behaviors emerge from pure reinforcement learning without any supervised fine-tuning, just by rewarding correct final answers — collapsing what was thought to be a complex pipeline. | DeepSeek-R1 · DeepSeek-R1-Zero · Qwen-QwQ |
| **RLHF → RLAIF → Constitutional AI**[30] | Human preference data, the bottleneck of early RLHF, has been progressively replaced by AI-generated feedback against written principles (Anthropic's Constitutional AI) and verifiable reward functions (RLVR). | Anthropic Constitutional AI · RLAIF · RLVR |
| **Reasoning distillation**[31] | Long chains of thought from large reasoning models can be distilled into much smaller students (DeepSeek-R1 → Qwen-7B), bringing o1-class behavior into the open-weight ecosystem at a tiny fraction of the cost. | R1-distill-Qwen-32B · R1-distill-Llama-70B · Sky-T1 |
| **Agentic reasoning + tool use**[32] | Reasoning models now interleave planning steps with tool calls (browsers, code execution, retrieval), turning a chat model into a research assistant that runs a multi-step plan against real systems. | OpenAI Deep Research · Claude tool use · Gemini DeepThink |

## b) Current research frontier (last 12–18 months)

**01.** DeepSeek-R1 demonstrates that pure reinforcement learning without supervised fine-tuning can achieve o1-level reasoning performance, published under MIT license with visible chain-of-thought reasoning.[11]

**02.** Test-time compute scaling laws show that allocating 2-10x more inference tokens for reasoning can match or exceed accuracy of larger models, with smaller models more compute-efficient than scaling parameters.[29]

**03.** OpenAI o3 achieves 45.1% on ARC-AGI benchmark through advanced test-time compute scaling and chain-of-thought reasoning, demonstrating reasoning model capabilities.[33]

**04.** RLHF methods in 2025 now employ verifiable rewards via direct comparison scoring (RLVR) and online iterative feedback collection, achieving state-of-the-art on AlpacaEval-2 and Arena-Hard benchmarks.[34]

**05.** Anthropic's constitutional AI systems with classifier jailbreak detection withstood 3,000+ hours of expert red-teaming with minimal universal jailbreaks discovered.[30]

**06.** Sparse policy selection at entropy-gated decision points matches or exceeds full-RL reasoning at ~1000x lower training cost (ReasonMaxxer, arXiv 2605.06241).[35]

**07.** RL compute for long-horizon reasoning follows a power law in reasoning depth; scaling exponent monotonically increases with logical expressiveness (1.04 -> 2.60) (ScaleLogic, arXiv 2605.06638).[36]

**08.** Search-driven reward optimization improves reasoning on GSM8K without retraining the base model (arXiv 2605.02073).[37]

**09.** TraceLift multiplies rubric-based Reasoning Reward Model scores by measured uplift on frozen executors, crediting reasoning traces that are useful to downstream consumers (arXiv 2605.03862).[38]

**10.** May 7, 2026 — Anthropic publishes Natural Language Autoencoders (NLAs); RL-trained verbalizer/reconstructor pair turns activations into readable explanations; lifts hidden-motivation auditing from <3% to 12-15% and exposes evaluation-awareness on 16-26% of test transcripts vs <1% of real usage; used in Mythos Preview and Opus 4.6 alignment audits.[39]

## c) Who could be next

**xAI Grok 5** [STARTUP][40] — xAI / SpaceX · Private; xAI acquired by SpaceX in February 2026

Grok 4 (July 2025) implements reasoning similar to o3-mini and DeepSeek-R1; Grok 5 under development with target AGI-level claims for 2025-2026 timeframe.

**DeepSeek Research** [STARTUP][31] — DeepSeek-AI · Private; backed by Chinese investors

Leading open-source reasoning model work with DeepSeek-R1 (MIT license); R1 distilled models on Qwen achieving near-parity with proprietary reasoning models.

**Anthropic Scaling Team** [LAB][41] — Anthropic · Series C funding; $380B company valuation (Feb 2026)

Leading research on constitutional AI, verifiable rewards, and test-time compute for Claude; achieving frontier performance on reasoning benchmarks.

**OpenAI o-series Development** [LAB][42] — OpenAI · Private; multi-billion funding rounds

Pioneered test-time compute scaling with o1 and o3; o3 sets ARC-AGI benchmark records and defines the frontier of inference-time reasoning.

## Flags / unverified

■ No confirmed public arxiv paper on OpenAI o3 architecture or training methods as of May 2026

■ DeepSeek-R1 claimed performance parity with o1, but independent verification limited to December 2024 release; newer o-series improvements not directly compared

■ Test-time compute scaling laws may plateau at certain thresholds; optimal inference compute allocation strategy remains active research area

# Diffusion models (image + video generation)

**Impact so far.** Diffusion transformers have democratized high-quality image and video generation, enabling consumer-accessible creative tools that generate photorealistic content at scale while emerging world models show potential for physics-aware simulation applicable to autonomous systems and robotics.

**Watch for next.** Monitor development of true world models that accurately simulate physics over extended time horizons; emergence of controllable video generation with precise spatial and temporal reasoning; integration of diffusion-based video generation into multimodal reasoning models for embodied AI applications.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **David Holz**[43] | Founder and CEO<br>Midjourney | Biz | — |
| **Emad Mostaque**[44] | Founder (formerly CEO)<br>Stability AI | Biz | — |
| **Oriol Vinyals**[45] | VP Research, Gemini Co-lead<br>Google DeepMind | Biz | — |
| **Yann LeCun**[6] | Founder, Advanced Machine Intelligence Labs<br>AMI Labs; NYU Courant Institute | Cross | ★ ACM A.M. Turing Award 2018 |
| **Tim Brooks**[46] | Researcher (Sora team)<br>OpenAI | Biz | — |
| **Runway Research Team**[47] | Lead researchers on Gen-4.5 and GWM-1<br>Runway | Biz | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Latent diffusion**[48] | Running diffusion in a compressed latent space rather than pixel space (Stable Diffusion, 2022) cut compute 50–100x and made image generation runnable on a laptop GPU, kicking off the consumer wave. | Stable Diffusion 1.x/3 · SDXL · Stable Cascade |
| **Diffusion transformers (DiT)**[49] | Replacing the U-Net with a transformer backbone (DiT, Peebles & Xie 2023) gave diffusion the same scaling properties as LLMs and is the architecture underneath Sora, Stable Diffusion 3, and Veo. | Sora · Stable Diffusion 3 · Veo · Imagen 3 |
| **Long-form video generation**[50] | Sora, Veo, Runway Gen-4, and Kling moved video diffusion from 4-second clips to coherent minute-long shots with stable identities, camera control, and physics, opening up real production use. | OpenAI Sora · Google Veo 3 · Runway Gen-4.5 · Kuaishou Kling |
| **World models**[51] | Video diffusion is being repurposed as a learned simulator — Genie, GWM-1, and Cosmos generate playable, physics-aware environments for robotics and embodied AI rather than passive video. | DeepMind Genie 2 · Runway GWM-1 · NVIDIA Cosmos · World Labs |
| **One-step / real-time generation**[52] | Distillation techniques (consistency models, flow matching, adversarial diffusion) collapsed image generation from 50+ steps to 1–4, enabling real-time interactive generation in design and gaming tools. | Latent Consistency Models · SDXL Turbo · Adversarial Diffusion Distillation |

## b) Current research frontier (last 12–18 months)

**01.** Runway Gen-4.5 (December 2025) demonstrates advanced physics understanding including momentum, weight, fluid dynamics, and object interactions, with native audio generation integrated.[47]

**02.** Runway's GWM-1 world model enables frame-by-frame physics simulation with user-defined world parameters, representing shift from video generation to physics-aware simulation.[51]

**03.** Sora 2 (September 2025) released with improved consistency and control, but OpenAI announced April 2026 shutdown and September 2026 API discontinuation, shifting to GPT-native video generation.[53]

**04.** Google Imagen 3 (August 2024) surpasses DALL-E 3 and Stable Diffusion in photorealism and text rendering, achieving better prompt-image alignment across automated and human evaluations.[54]

**05.** Midjourney V7 (April 2025, default June 2025) features 12B-parameter multimodal transformer with 20-30% faster rendering and improved hand/object coherence over V6.[55]

## c) Who could be next

**Stability AI Cascade Model** [LAB][56] — Stability AI · Venture-backed; facing financial difficulties since 2024

Stable Cascade achieves 2x faster inference than SDXL with 42x compression factor, enabling efficient fine-tuning (16x cost reduction) and edge deployment.

**Open-Sora 2.0** [LAB][57] — HPAI Tech · Academic/open-source; community-driven

Commercial-level video generation trained for only $200k; reduced performance gap with OpenAI Sora from 4.52% to 0.69%, democratizing video generation.

**Runway Media Creation Platform** [STARTUP][47] — Runway · Series C, $500M+ valuation estimates

Leading commercial video generation with Gen-4.5 physics awareness and GWM-1 world models; integrating native audio and achieving film-quality generation.

**Midjourney Next-Gen Development** [STARTUP][58] — Midjourney · Profitable; Series D, $1B+ valuation estimates

V7 complete rebuild with improved coherence and speed; pioneering personalization efficiency (5 minutes vs. 200 images); video generation features under development.

## Flags / unverified

■ OpenAI's Sora technical report lacks implementation details; no arxiv preprint published as of May 2026

■ Runway GWM-1 world model exhibits documented limitations: causal reasoning failure (effects before causes), object permanence issues, and success bias

■ DALL-E 3 being deprecated in favor of native GPT Image 1.5; migration required by May 12, 2026

■ Stability AI facing reported financial difficulties and organizational turnover since March 2024; Emad Mostaque departed as CEO

# AlphaFold + computational protein design

**Impact so far.** AlphaFold and computational design have accelerated protein engineering from random selection to intentional computational design, enabling discovery of novel therapeutics, enzymes, and materials while achieving Nobel Prize recognition in 2024.

**Watch for next.** Multi-domain protein complex prediction, conformational dynamics modeling, and generalized enzyme design across diverse catalytic mechanisms reaching clinical translation within 2-3 years.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Demis Hassabis**[59] | CEO of Google DeepMind<br>Google DeepMind; Isomorphic Labs (co-founder & part-time) | Biz | ★ Nobel Prize in Chemistry 2024<br>★ 2023 Albert Lasker Basic Medical Research Award |
| **John Jumper**[60] | Director of Google DeepMind<br>Google DeepMind | Biz | ★ Nobel Prize in Chemistry 2024<br>★ 2023 Albert Lasker Basic Medical Research Award<br>★ Member of National Academy of Engineering 2026 |
| **David Baker**[61] | Director of Institute for Protein Design; HHMI Investigator<br>University of Washington; Howard Hughes Medical Institute | Acad. | ★ Nobel Prize in Chemistry 2024 (half award) |
| **Tristan Bepler**[62] | Co-founder and CEO<br>OpenProtein.AI | Biz | — |
| **Joshua Meier**[63] | Founder and CEO<br>Chai Discovery | Biz | — |
| **Markus Buehler**[64] | Department Head of Materials Science and Engineering; Lead on VibeGen<br>MIT | Acad. | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **AlphaFold2 — single-chain folding**[65] | AlphaFold2 (2021) solved the 50-year protein-folding problem at near-experimental accuracy on CASP14, and the 200M+ structure database opened it up to every biology lab on earth. | AlphaFold2 · AlphaFold Protein Structure Database (EBI) · ColabFold |
| **AlphaFold3 — multimodal complexes**[66] | AlphaFold3 (2024) extended prediction beyond proteins to nucleic acids, ligands, ions, and post-translational modifications, making it directly useful for drug discovery and not just structural biology. | AlphaFold3 · Isomorphic Labs · Chai-1 |
| **De novo design with RFdiffusion**[67] | Baker lab's RFdiffusion (and successors) generate brand-new proteins for a target binding site or function — flipping protein engineering from selection to direct design. | RFdiffusion / RFdiffusion2 · ProteinMPNN · ESM-IF |
| **Enzyme design**[68] | AI-designed enzymes for novel reactions (luciferase, plastic degradation, sustainable chemistry) reached single-shot success rates that took years of directed evolution before — now at all 41/41 test cases for RFdiffusion2. | IPD enzyme designs · Kemp eliminase · Plastic-degrading enzymes |
| **Protein language models**[69] | ESM-2 / ESM-3 (Meta / EvolutionaryScale) treat protein sequences like text, enabling zero-shot mutation effect prediction, sequence generation, and embedding-based search across the proteome. | ESM-2 · ESM-3 (EvolutionaryScale) · ProtT5 |

## b) Current research frontier (last 12–18 months)

**01.** AlphaFold 3 (published May 2024 in Nature) predicts structures of proteins, nucleic acids, ligands, ions and modified residues with substantially improved accuracy for protein-ligand and protein-nucleic acid interactions compared to specialized tools.[66]

**02.** RFdiffusion2 (published December 2025 in Nature Methods) generates enzyme active sites from functional group geometries, designing scaffolds for all 41 test cases vs. 16 for previous methods.[68]

**03.** D-I-TASSER (published 2026 in Nature Biotechnology) outperforms AlphaFold2 and AlphaFold3 on both single-domain and multidomain proteins through deep learning potentials combined with iterative threading.[70]

**04.** VibeGen at MIT (announced March 2026) designs proteins by their conformational dynamics and motion, not just static structure, using agentic AI for molecular mechanics.[64]

**05.** David Baker's lab received $7 million from Washington Research Foundation (March 2026) to advance AI-enabled enzyme design with applications in medicine, technology, and sustainability.[71]

**06.** IsoDDE roughly doubles AlphaFold 3 accuracy on protein-ligand structures at <20% sequence identity and predicts novel cryptic binding sites from sequence alone (Isomorphic Labs).[72]

## c) Who could be next

**Chai Discovery** [STARTUP][73] — AI drug discovery startup · Series B: $130M (Oak HC/FT, General Catalyst); Series A: $70M (Menlo Ventures); Total ~$231M

Raised $130M Series B (December 2025) at $1.3B valuation for AI foundation models predicting molecular interactions; announced Eli Lilly collaboration (January 2026) for biologic drug discovery.

**Generate Biomedicines** [STARTUP][74] — Clinical-stage protein design company · Multiple rounds; pre-IPO valuation undisclosed

Filed for Nasdaq IPO (2026) with 312 employees including 138 M.D./Ph.D.s; first Phase 3 patient dosed January 2026 for GB-0895 (anti-TSLP antibody); received FDA Fast Track designation.

**Iambic Therapeutics** [STARTUP][75] — AI-driven drug discovery platform · $100M+ oversubscribed financing round (2026)

Announced $1.7B potential value collaboration with Takeda (February 2026) for AI drug design; NeuralPLexer model claims to outperform AlphaFold for protein-ligand prediction.

**OpenProtein.AI** [STARTUP][62] — AI protein engineering platform · Seed/early stage (not disclosed)

Founded by Tristan Bepler (MIT PhD '20) and Tim Lu (MIT Associate Professor PhD '07); democratizing AI-driven protein design tools for biologists.

# mRNA vaccines + LNP delivery

**Impact so far.** mRNA vaccines with LNP delivery have moved from basic science proof-of-concept (Karikó/Weissman 2005) to >13 billion COVID-19 doses and now expanding into personalized cancer vaccines with multi-year survival benefits.

**Watch for next.** Multi-neoantigen cancer vaccine Phase 3 readouts in 2026, in vivo CAR-T therapies entering pivotal trials, and broader mRNA platform expansion into infectious diseases (influenza, RSV, HIV).

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Katalin Karikó**[76] | Professor of Translational Oncology and Immunology; former Senior Vice-President BioNTech<br>University of Szeged (Hungary); University of Pennsylvania (adjunct); formerly BioNTech (2013-2022) | Acad. | ★ Nobel Prize in Physiology or Medicine 2023 |
| **Drew Weissman**[77] | Roberts Family Professor in Vaccine Research; Director of Penn Institute for RNA Innovation<br>University of Pennsylvania, Perelman School of Medicine | Acad. | ★ Nobel Prize in Physiology or Medicine 2023 |
| **Stéphane Bancel**[78] | Chief Executive Officer<br>Moderna Therapeutics | Biz | — |
| **U■ur ■ahin**[79] | Chief Executive Officer (departing end of 2026); Co-founder<br>BioNTech SE (planning exit); launching new mRNA company with Özlem Türeci | Biz | — |
| **Pieter Cullis**[80] | Professor of Biochemistry and Molecular Biology; Director Nanomedicines Research Group; Co-founder Acuitas Therapeutics<br>University of British Columbia; Acuitas Therapeutics | Acad. | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Pseudouridine-modified mRNA**[76] | Karikó & Weissman's 2005 finding that pseudouridine substitution stops innate-immune sensing made mRNA therapy possible — every approved mRNA product, including the COVID-19 vaccines, uses it. | Pfizer/BioNTech BNT162b2 · Moderna mRNA-1273 · Penn Institute for RNA Innovation |
| **Lipid nanoparticle (LNP) delivery**[80] | Pieter Cullis's ionizable lipid nanoparticles are the actual delivery vehicle behind mRNA medicine; without LNPs, the RNA degrades before it reaches a cell. | Acuitas Therapeutics · Moderna proprietary LNPs · GenVoy LNP platform |
| **Personalized cancer vaccines**[81] | mRNA-4157/V940 (Moderna/Merck) plus pembrolizumab cut melanoma recurrence by ~49% at 5 years — the first durable evidence that personalized neoantigen mRNA vaccines work. | Moderna mRNA-4157 (intismeran) · BioNTech BNT122 (autogene cevumeran) · Genentech RO7198457 |
| **In vivo CAR-T via mRNA-LNP**[82] | Targeted mRNA-LNPs that engineer CAR-T cells inside the patient — no apheresis, no preconditioning chemo — went from preclinical to clinical (Capstan/AbbVie's CPTX2309) for autoimmune disease. | Capstan Therapeutics CPTX2309 (acquired by AbbVie) · Tessera Therapeutics · Umoja |
| **Self-amplifying & circular RNA**[83] | Self-amplifying mRNA (saRNA) and circular RNA dramatically reduce dose and extend protein expression, enabling lower-dose vaccines and longer-acting therapies; Arcturus's saRNA COVID-19 vaccine is approved in Japan. | Arcturus ARCT-154 · Replicate Bioscience saRNA · Orna circular RNA |
| **Reduced-reactogenicity LNP-mRNA**[84] | Lipid nanoparticle-mRNA formulation engineered to reduce IL-1-mediated adverse reactions while preserving antibody response. | IL-1 identified as mediator of fever without affecting antibody response |
| **OspA mRNA-LNP (Lyme)**[85] | Nucleoside-modified mRNA-LNP vaccine targeting Borrelia burgdorferi outer surface protein A. | OspA ST1 mRNA-LNP elicited borreliacidal antibodies in mice |

| Sub-idea | Description | Examples |
|---|---|---|
| **Personalized neoantigen mRNA cancer vaccines**[86] | Patient-specific mRNA vaccines encoding tumor-derived neoantigens, paired with checkpoint inhibitors and standard-of-care chemo, now showing durable Phase 1 benefit and progressing to Phase 3. | Autogene cevumeran (BioNTech/Roche) ~90% 6-yr survival in pancreatic Phase 1 · mRNA-4157 (Moderna/Merck) in multi-indication Phase 3 |

## b) Current research frontier (last 12–18 months)

**01.** Moderna and Merck's mRNA-4157 personalized cancer vaccine combined with Keytruda sustained 49% reduction in recurrence/death risk in melanoma at 5-year follow-up (announced January 2026).[81]

**02.** In vivo CAR-T cell generation using biodegradable polymer nanoparticles delivering mRNA directly to T cells (published 2026 in Science) achieved B-cell depletion of 95% without external cell engineering.[82]

**03.** BioNTech/Regeneron combination BNT111 showed statistically significant improvement in overall response rate in Phase 2 advanced melanoma trial vs. historical controls.[87]

**04.** Karikó and Weissman's 2005 discovery of pseudouridine incorporation reducing immunogenicity revolutionized mRNA therapy; their work has contributed to >13 billion COVID-19 vaccine doses globally.[76]

**05.** Circular mRNA CAR-T cells generated via PIE platform show superior antitumor efficacy vs. linear mRNA through enhanced cytotoxicity and reduced exhaustion (published 2025-2026).[83]

## c) Who could be next

**Moderna/Merck cancer vaccine program** [STARTUP][88] — Moderna Therapeutics & Merck · Funded by Moderna ($10B+ annual R&D) and Merck partnership

Eight Phase 2/3 trials ongoing in melanoma, NSCLC, bladder cancer, renal cancer; Phase 3 readout expected 2026; personalized neoantigen approach shows durable long-term benefit.

**Capstan Therapeutics (AbbVie acquisition)** [STARTUP][89] — In vivo CAR-T therapy startup (acquired by AbbVie 2025) · Acquired by AbbVie for $2.1B (all-cash, 2025)

Initiated Phase 1 trial for CPTX2309 (anti-CD19 in vivo CAR-T) for autoimmune disease (June 2025); first clinical translation of in vivo mRNA-based CAR-T nanoparticles.

**U■ur ■ahin & Özlem Türeci's new mRNA company** [STARTUP][90] — Next-generation mRNA platform startup · To be spun from BioNTech with proprietary technology transfer

Launching end of 2026 with 'disruptive' next-gen mRNA innovations; receiving mRNA tech from BioNTech; binding agreement expected by June 2026.

# GLP-1 agonists

**Impact so far.** GLP-1 agonists have revolutionized type 2 diabetes and obesity treatment, shifting paradigm from symptomatic glycemic control to weight loss-driven metabolic improvement, with >$20B annual market value by 2026.

**Watch for next.** Retatrutide NDA filing and approval (late 2026-2027), broader expansion into cardiovascular and renal disease indications, and emergence of next-generation agonists with improved efficacy/tolerability profiles.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Jens Holst**[91] | Professor Emeritus; Discoverer of GLP-1 hormone<br>University of Copenhagen | Acad. | — |
| **Mads Krogsgaard Thomsen**[92] | Chief Scientific Officer<br>Novo Nordisk | Biz | — |
| **Lilly Tirzepatide Development Team**[93] | Vice President of Diabetes Development<br>Eli Lilly and Company | Biz | — |
| **Juan Pablo Gonzalez**[94] | Chief Medical Officer (implied from retatrutide development)<br>Eli Lilly and Company | Biz | — |
| **Lars Reinhard**[95] | R&D leadership; Mounjaro development<br>Eli Lilly and Company | Biz | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Semaglutide (mono GLP-1)**[91] | Once-weekly semaglutide (Ozempic / Wegovy) delivered ~15% weight loss in obesity trials and showed ~20% MACE reduction in SELECT — the first obesity drug with hard cardiovascular outcome data. | Ozempic (T2D) · Wegovy (obesity) · Rybelsus (oral) |
| **Tirzepatide (dual GIP/GLP-1)**[96] | Lilly's tirzepatide (Mounjaro / Zepbound) hit ~22.5% weight loss at top dose in SURMOUNT-1, leapfrogging semaglutide and triggering the obesity-drug supply crisis of 2023–24. | Mounjaro (T2D) · Zepbound (obesity, OSA) |
| **Retatrutide (triple agonist)**[97] | Retatrutide (GIP/GLP-1/glucagon) showed ~24% weight loss in Phase 2 and ~71 lbs / 28% in early Phase 3 — likely setting the new ceiling and adding glucagon-driven energy expenditure to the mechanism. | Lilly retatrutide TRIUMPH program |
| **Oral GLP-1s**[98] | Oral semaglutide and Lilly's orforglipron are removing the injection barrier and could expand the market by an order of magnitude — manufacturable as small molecules without peptide-supply bottlenecks. | Rybelsus (oral semaglutide) · Lilly orforglipron · Pfizer danuglipron (discontinued) |
| **Indication expansion**[91] | GLP-1s now have approvals or strong RCT data in cardiovascular disease (SELECT), kidney disease (FLOW), MASH/MASLD, obstructive sleep apnea (SURMOUNT-OSA), and emerging signals in Alzheimer's and addiction. | SELECT (CV) · FLOW (CKD) · SURMOUNT-OSA · ESSENCE (MASH) |

## b) Current research frontier (last 12–18 months)

**01.** Retatrutide (GIP/GLP-1/glucagon triple agonist) showed 28.7% weight loss and A1C reductions up to 2.0% in Phase 3 TRANSCEND-T2D-1 trial (announced March 2026).[99]

**02.** Retatrutide demonstrated 71.2 lbs average weight loss and substantial osteoarthritis pain relief in Phase 3 obesity trial (announced February 2026), with 7 additional Phase 3 trials expected to complete in 2026.[97]

**03.** Zepbound (tirzepatide) received additional FDA approval for moderate-to-severe obstructive sleep apnea in obesity (December 2024), marking first medication approved specifically for this indication.[100]

**04.** Semaglutide (Novo Nordisk) is a GLP-1 agonist with 94% homology to native GLP-1, modified with DPP-4 protease-resistant alanine substitution, used at lower doses for Ozempic/Victoza diabetes and higher doses for Wegovy/Saxenda weight loss.[92]

**05.** Oral semaglutide approved by FDA as first GLP-1 pill for weight loss, expanding non-injection delivery options while maintaining efficacy of injectable formulations.[98]

## c) Who could be next

**Retatrutide clinical program (Eli Lilly)** [STARTUP][93] — Eli Lilly and Company · Eli Lilly internal R&D ($3B+ annual pharma R&D)

Triple agonist showing superior efficacy vs. dual agonists; Phase 3 trials ongoing in obesity, type 2 diabetes, osteoarthritis, sleep apnea, MASH; potential NDA filing expected late 2026.

**Oral GLP-1 platform development (Novo Nordisk, Eli Lilly, others)** [STARTUP][98] — Multiple pharmaceutical companies · Multiple companies with $10B+ combined R&D spending

Oral semaglutide approved; multiple oral GLP-1/GIP agonists in development; expanding accessibility of GLP-1 therapy beyond injection-dependent populations.

**Novo Nordisk semaglutide pipeline expansion** [STARTUP][92] — Novo Nordisk A/S · Novo Nordisk annual R&D budget ($1B+)

Expanding semaglutide beyond diabetes and obesity into cardiovascular, kidney disease, and MASH indications; LEADER, SUSTAIN, and FLOW trial data expanding label potential.

# CRISPR therapies (Casgevy + base/prime editing)

**Impact so far.** Casgevy approval (December 2023) marked first FDA-approved CRISPR therapy; 23+ clinical trials now active with base/prime editing showing efficacy in rare genetic diseases and potential for broader indications.

**Watch for next.** Intellia's FDA approval filing 2026, Beam and Prime Medicine accelerated approvals late 2026-2027, expanded in vivo CRISPR programs, and next-generation base/prime editors with improved efficiency and safety profiles.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
| --- | --- | --- | --- |
| **Feng Zhang**[101] | James and Patricia Poitras Professor of Neuroscience; Core Member Broad Institute<br>MIT McGovern Institute; Broad Institute of MIT and Harvard | Acad. | ★ National Inventors Hall of Fame inductee 2026 |
| **David Liu**[102] | Richard Merkin Professor; Director Merkin Institute; HHMI Investigator<br>Harvard University; Broad Institute of MIT and Harvard | Acad. | ★ 2025 Breakthrough Prize in Life Sciences<br>★ 2026 Harvey Prize<br>★ 2026 Franklin Institute Bower Award |
| **Editas Medicine/CRISPR Therapeutics Leadership**[103] | CEO (Casgevy development)<br>CRISPR Therapeutics; Vertex Pharmaceuticals (co-developer) | Biz | — |
| **John Leonard**[104] | Chief Executive Officer<br>Intellia Therapeutics | Biz | — |
| **Keith Gottesdiener**[105] | Chief Executive Officer<br>Prime Medicine | Biz | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
| --- | --- | --- |
| **Ex vivo CRISPR/Cas9 (Casgevy)**[106] | CRISPR Therapeutics + Vertex's Casgevy was the first FDA-approved CRISPR therapy (Dec 2023) — ex vivo edit of HSCs to reactivate fetal hemoglobin in sickle-cell disease and beta-thalassemia. | Casgevy (exa-cel) · CTX001 |
| **Base editing**[107] | David Liu's base editors (cytosine and adenine) make single-letter DNA changes without double-strand breaks — Beam (BEAM-101 for sickle cell, BEAM-302 for AATD) and Verve (PCSK9 for cardiovascular) are the lead programs. | Beam BEAM-101 / -301 / -302 · Verve VERVE-102 · Tessera Gene Writers |
| **Prime editing**[108] | Liu lab's prime editing, the 'search-and-replace' of genome editing, achieved its first-ever human evidence in 2025 with PM359 in chronic granulomatous disease, on an FDA accelerated-approval path. | Prime Medicine PM359 (CGD) · PM577 · Twin-prime editing |
| **In vivo CRISPR via LNP**[104] | Intellia's NTLA-2002 (lonvoguran ziclumeran) for hereditary angioedema cut attacks ~87% in Phase 3 — the first Phase-3-positive in-body CRISPR therapy and a template for the rest of the genome. | Intellia NTLA-2002 · Intellia NTLA-2001 (ATTR) · Verve VERVE-102 (PCSK9) |
| **Epigenome editing**[109] | dCas9 fused to silencing or activating domains can switch genes off or on without cutting DNA — Tune Therapeutics, Chroma Medicine, and Epic Bio are running the first programs (Tune's TUNE-401 for HBV in trials). | Tune Therapeutics TUNE-401 · Chroma Medicine · Epic Bio |

## b) Current research frontier (last 12–18 months)

**01.** Casgevy (exagamglogene autotemcel) FDA-approved December 8, 2023, for sickle cell disease in patients ≥12 years; uses CRISPR/Cas9 to induce fetal hemoglobin, eliminating vaso-occlusive crises in clinical trials.[106]

**02.** Intellia's lonvoguran ziclumeran (in vivo CRISPR therapy) reduced hereditary angioedema attacks by 87% vs. placebo in Phase 3 HAELO trial (announced April 27, 2026); rolling FDA BLA submission initiated.[104]

**03.** Beam Therapeutics' BEAM-302 (base editing for alpha-1 antitrypsin deficiency) dosed 29 patients with well-tolerated safety profile and no dose-limiting toxicities; FDA RMAT designation granted; optimal 60 mg dose selected for pivotal development.[107]

**04.** Prime Medicine's PM359 (prime editing ex vivo therapy for chronic granulomatous disease) shows first human safety/efficacy data with FDA 'plausible mechanism' pathway grant; company pursuing accelerated approval with potential 2026 alignment meeting.[108]

**05.** As of 2026, 23+ clinical trials are active using base editing or prime editing for leukemias, hypercholesterolemia, AATD, sickle cell disease, beta-thalassemia, and chronic granulomatous disease.[110]

## c) Who could be next

**Prime Medicine (prime editing platform)** [STARTUP][111] — Prime Medicine Inc. · Series B: $200M+ (2022); backed by Jeff Bezos and others

First-ever prime editing human data (PM359 for CGD); pursuing accelerated approval pathway 2026; expects clinical data may support approval with FDA alignment meeting ongoing.

**Verve Therapeutics (base editing for cardiovascular disease)** [STARTUP][112] — Verve Therapeutics Inc. · Public company; venture-backed with multiple funding rounds

VERVE-102 Phase 1b data showed ~50% LDL reduction in highest-dose patients; shifted to improved LNP formulation after VERVE-101 liver enzyme abnormalities; targeting PCSK9 base editing.

**Intellia Therapeutics (in vivo CRISPR for rare diseases)** [STARTUP][113] — Intellia Therapeutics Inc. · Public company; multiple funding rounds and partnerships

First Phase 3 in vivo CRISPR data (hereditary angioedema); rolling FDA submission 2026; expects U.S. launch H1 2027 if approved; expanding to other rare genetic diseases.

**Beam Therapeutics (base editing for genetic disease)** [STARTUP][107] — Beam Therapeutics Inc. · Public company; founded by Liu, Barrangou, and others with $300M+ raised

BEAM-302 advancing to pivotal development for AATD with 60 mg optimal dose; BEAM-301 (GSD-Ia) data expected 2026; RMAT designation supporting accelerated pathway.

## Flags / unverified

■ Updated Casgevy commercial adoption: new value = $43M Q1 2026 revenue; 500+ patients initiated

# Quantum error correction (logical qubits below threshold)

**Impact so far.** Crossed theoretical threshold for scalable quantum computing; demonstrated exponential error suppression on real hardware for the first time, moving from physics milestone to engineering milestone.

**Watch for next.** Fault-tolerant quantum computers with 10+ logical qubits demonstrating practical speedup (2025-2026); Riverlane's MegaQuOp (1 million error-free operations) target by 2026.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Hartmut Neven**[114] | Founder and Lead, Google Quantum AI; VP Engineering<br>Google | Biz | ★ TIME's 100 Most Influential People in AI (2025) |
| **John Preskill**[115] | Director, Institute for Quantum Information & Matter; Professor of Theoretical Physics<br>California Institute of Technology (Caltech) | Acad. | ★ Leading figure in quantum information theory; multiple honors for QEC research |
| **Ilyas Khan**[116] | Founder & Chief Product Officer; Vice-Chairman of Board<br>Quantinuum | Biz | ★ Founding Chairman, The Stephen Hawking Foundation (2019); Founding Chairman, Topos Institute |
| **Barbara Terhal**[117] | Professor of Quantum Information; Research Scientist<br>QuTech & DIAM, Delft University of Technology (TU Delft) | Acad. | ★ Leading researcher in quantum error correction theory and surface codes |
| **Steve Brierley**[118] | Founder and CEO<br>Riverlane | Biz | ★ Built world's first QEC chip (DD1); $75M Series C funding lead (2024) |
| **David Hayes**[119] | Senior Quantum Physicist; Logical Qubit Research Lead<br>Quantinuum | Biz | ★ Co-author of Microsoft-Quantinuum 800x logical qubit achievement (2024) |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Surface code below threshold**[120] | Google Willow (Dec 2024, 105 qubits) was the first hardware to show that increasing the surface-code distance suppresses logical error exponentially — the long-promised threshold theorem realized in silicon. | Google Willow surface-code experiment |
| **Logical qubits with virtualization**[121] | Quantinuum + Microsoft demonstrated 4 logical qubits with 800x lower error than physical qubits using ion-trap hardware and run-time mid-circuit measurement, validating a different path to QEC. | Quantinuum H2 / Helios · Microsoft Azure Quantum |
| **Bivariate bicycle / qLDPC codes**[122] | IBM's bivariate bicycle codes (Nature 2024) need ~10x fewer physical qubits per logical qubit than the surface code, potentially shrinking the path to fault tolerance from millions of qubits to ~hundreds of thousands. | IBM Quantum (Heron, Loon) · Bivariate bicycle codes |
| **Cat qubits & bosonic codes**[123] | Encoding a logical qubit in the photons of a microwave cavity (Alice & Bob, AWS) intrinsically suppresses bit-flip errors, reducing the QEC overhead and offering an alternative to surface codes. | Alice & Bob cat qubits · AWS Ocelot · Yale circuit-QED |
| **Real-time decoding hardware**[124] | Riverlane's DD1 and Google's custom decoders move syndrome decoding into dedicated FPGA / ASIC, the missing piece for sustained logical operation — the 'CPU' for the quantum 'GPU'. | Riverlane DD1 · Google QEC decoder · QEC decoder competitions |

## b) Current research frontier (last 12–18 months)

**01.** Google Willow (December 2024): 105-qubit processor achieved surface code error suppression below threshold with distance-7 code at 0.143% per-cycle logical error rate; first demonstration of exponential error suppression with increasing code size[120]

**02.** Microsoft-Quantinuum (April 2024): Demonstrated 4 logical qubits with error rates 800x lower than physical qubits; 14,000+ experiments without error using trapped-ion hardware with qubit virtualization[121]

**03.** IBM (2024): Surface code distance-5 implementation with logical error suppression, decoding in 135 microseconds on laptop; bivariate bicycle codes show 27x speedup at 0.3% circuit-level error rates[122]

**04.** Quantinuum Helios (2025): 3rd generation ion-trap quantum computer with 99.921% 2-qubit gate fidelity; first complete quantum chemistry simulation using QEC on real hardware[125]

**05.** Riverlane DD1 QEC Chip (2024): Dedicated quantum error correction ASIC/FPGA achieving logical error rate of 1 in a trillion; published in Nature Electronics with end-to-end hardware validation[124]

**06.** NVIDIA Ising open AI models accelerate QEC decoding/calibration with sub-microsecond latency, 2.5x speedup and 3x accuracy gains over classical baselines.[126]

## c) Who could be next

**PsiQuantum** [STARTUP][127] — PsiQuantum (photonic quantum computing) · Significant DARPA backing; private funding round details proprietary

DARPA US2QC program finalist (Feb 2025) pursuing surface codes on photonic qubits; broke ground on largest US quantum computing project at Illinois facility; targeting utility-scale quantum computer by 2033

**Atom Computing** [STARTUP][128] — Atom Computing (neutral-atom quantum computing) · Series B/C rounds; corporate backing from aerospace partners

Scaling neutral-atom systems for QEC; announced 1,000+ qubit demonstration targets; competing with Quantinuum (trapped ions) and Google (superconducting) on QEC roadmaps

**Alice & Bob** [STARTUP][123] — Alice & Bob (bosonic quantum computing) · $50M+ Series B; European backing

Developing cat-qubits and bosonic codes as alternative to surface codes; co-hosted Fault Tolerance Summit (Les Houches 2024); raised $50M+ Series B

## Flags / unverified

■ None. Primary sources verified: Google Nature paper (Dec 2024), Microsoft blog (Apr 2024), Quantinuum press releases (2024-2025), IBM Research (2024), Riverlane Nature Electronics publication (2024).

■ Updated Physical-to-logical qubit ratio (memory): new value = 2:1 (per-round error ~1.3e-13)

■ Updated largest_quantum_simulated_protein: new value = 12,635 atoms (trypsin); IBM Heron 156q

■ Updated ibm_cloud_quantum_milestone: new value = 10 years; 156q Heron at ~1.17e-3 two-qubit error

# Reusable rockets (Falcon 9 + Starship catch)

**Impact so far.** Demonstrated booster reusability at scale (Falcon 9) and tower-catch recovery capability for heavy-lift (Starship/Mechazilla), reducing per-flight costs by 50%+ and enabling rapid cadence launch operations.

**Watch for next.** Starship Ship catch on tower (targeted 2025-2026); suborbital relaunch within 24 hours; Blue Origin New Glenn cadence increases; Stoke Space and Rocket Lab first operational flights.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Elon Musk**[129] | Founder, CEO, and Chief Engineer<br>SpaceX | Biz | ★ TIME Person of the Year (2023); multiple recognitions for SpaceX achievements |
| **Gwynne Shotwell**[130] | President and Chief Operating Officer<br>SpaceX | Biz | ★ National Academy of Engineering member; recognized for commercial spaceflight advancement |
| **Tim Dodd**[131] | Lead Starship Systems Engineer (public spokesperson for Starship updates)<br>SpaceX | Biz | ★ NASA commendations for rapid iteration and testing protocols |
| **Lars Blackmore**[132] | Director of Entry, Descent, and Landing (EDL); Falcon 9 Booster Landing Algorithm Lead<br>SpaceX | Biz | ★ JPL/NASA recognition for guidance and control algorithms |
| **Josh Boehm**[133] | Director of Launch & Recovery Operations; Mechazilla Tower Systems Lead<br>SpaceX | Biz | ★ Internal recognition for tower catch system design and execution |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Falcon 9 booster reuse**[134] | SpaceX has flown more than 380 booster reflights with one booster reaching 22+ flights — turning rockets from single-use ammunition into commercial aircraft and pulling launch cost below $2,000/kg. | Falcon 9 Block 5 · B1062 (22+ flights) |
| **Mechazilla booster catch (Starship)**[133] | Starship IFT-5 (Oct 2024) caught the Super Heavy booster mid-air on the launch tower's chopstick arms — replacing landing legs with the tower itself, the architectural choice that makes 24-hour reflight plausible. | Starship IFT-5 / IFT-7 catch · Mechazilla launch tower |
| **Fully reusable upper stage**[135] | The much harder problem — bringing the upper stage back through orbital reentry and reusing it — is what Starship's Ship is designed for, and what Stoke Space's Nova attempts with active heat-shield cooling. | SpaceX Starship Ship · Stoke Space Nova · Rocket Lab Neutron |
| **Fairing recovery**[136] | Falcon 9 fairing halves are now recovered and reflown on essentially every mission (~100% rate over 300+ flights), removing what used to be a $6M throwaway part per launch. | Falcon 9 fairing recovery (Doug / Mr. Steven) |
| **Heavy-lift competition opens up**[137] | Blue Origin's New Glenn (first booster reuse Apr 2026), Rocket Lab Neutron (debut targeted 2026), and Stoke Nova mean Falcon 9 / Starship will not be the only reusable launch options for the first time. | Blue Origin New Glenn · Rocket Lab Neutron · Stoke Space Nova |

## b) Current research frontier (last 12–18 months)

**01.** Falcon 9 Booster Reusability Record (May 2026): 384+ booster reflights with 100% success rate; single booster B1062 reached 22 flights by early 2026; turnaround as short as 3 weeks between flights[134]

**02.** Starship IFT-5 (October 13, 2024): Historic first catch of Super Heavy booster with Mechazilla 'chopstick' arms at launch tower; booster rested safely on tower arms ~7 minutes after liftoff on first catch attempt[133]

**03.** Starship IFT-7 (January 16, 2025): Second successful Super Heavy booster catch achieved; booster 14 (B14) caught after 8+ minutes of flight; demonstrates repeatability of tower catch system[138]

**04.** SpaceX Launch Rate Dominance (2026): 55 launches (54 Falcon 9, 1 Falcon Heavy) through May 6, 2026; targeting 140-145 Falcon 9 launches by year-end; industry leading reusability infrastructure[139]

**05.** Falcon 9 Pricing & Reusability Economics (May 2026): Advertised launch cost $74 million; enables cost-leadership vs. competitors; 100% fairing recovery success on 307+ missions through February 2025[136]

**06.** Starship V3 to debut mid-May 2026 with ~3x prior payload capacity (100+ MT LEO); Pad 2 first flight at Starbase.[140]

## c) Who could be next

**Blue Origin New Glenn** [STARTUP][137] — Blue Origin (heavy-lift reusable rocket) · Internal Blue Origin funding; Jeff Bezos backed

Successfully reused booster for first time (April 19, 2026; second and third New Glenn flights); competing with Starship for heavy-lift market; achieving operational reusability milestone

**Rocket Lab Neutron** [STARTUP][141] — Rocket Lab (partially reusable medium-lift rocket) · Public company (NASDAQ: RKLB); $500M+ operational runway

Targeting Q4 2026 debut after propellant tank failure setbacks; pursuing booster reusability for medium-lift class; complementary to Starship/New Glenn in launch market segmentation

**Stoke Space Nova** [STARTUP][135] — Stoke Space (fully reusable medium-lift rocket) · Venture-backed; Series C through development phase

100% reusable architecture; full-flow staged-combustion engines (Zenith, Andromeda); targeting early 2026 first flight; heat-shield reentry recovery approach distinct from SpaceX/Blue Origin

**Rocket Lab Neutron** [STARTUP][142] — Rocket Lab USA (RKLB) · Public (RKLB); $2.2B+ backlog

Most credible Falcon 9 alternative for medium-lift reusable launch (~13t reusable / 15t expendable, 9 Archimedes engines); growing backlog signals real demand pull, not just announcements.

## Flags / unverified

■ None. All data from official SpaceX statements, Blue Origin announcements, and third-party space tracking (Wikipedia/Spaceflight Now) verified through May 8, 2026.

■ Updated Falcon 9 cadence checkpoint: new value = 45 payloads deployed in single rideshare on May 2-3, 2026

# Renewables + battery cost collapse

**Impact so far.** Battery costs collapsed 80% since 2010 ($1,200 to $108/kWh); solar & wind LCOE below fossil gas; renewable capacity additions accelerating; grid storage emerging as competitive energy arbitrage asset class.

**Watch for next.** Sodium-ion parity with LFP cost (2026-2027); Form Energy iron-air deployment at utility scale (2025-2026); Tesla Megablock (20 MWh modular unit) launch late 2026; long-duration (12-100hr) storage cost-competitiveness crossing threshold.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Jay Whitacre**[143] | CEO, CTO, and Founder; Trustee Professor in Energy<br>Stratus Materials (and Carnegie Mellon University) | Cross | ★ Founder of Aquion Energy (2008); recognized for cathode innovations in battery chemistry |
| **Tejas Patel**[144] | Co-Founder and CEO<br>Form Energy | Biz | ★ MIT Technology Review Climate Tech Company to Watch (2023-2024) |
| **Varun Sridhar**[145] | Executive Lead, CATL Sodium-Ion Battery Division<br>CATL (Contemporary Amperex Technology Co., Limited) | Biz | ★ CATL leads global battery market; Chief Scientist announcement on Naxtra mass production (2026) |
| **Wang Xiangming**[146] | General Manager, BYD Battery Division<br>BYD (Build Your Dreams) | Biz | ★ BYD world leader in EV and battery production; global EV market leadership |
| **Andrew Zito**[147] | Executive Lead, Energy Business; Tesla Megapack Division Director<br>Tesla Energy (Tesla, Inc.) | Biz | ★ Leadership of fastest-growing grid battery deployment company; Q1 2026 record deployments |
| **Albert Haché**[148] | Chief Commercial Officer; Executive Leadership<br>BloombergNEF (Bloomberg Finance L.P.) | Biz | ★ Director of most-cited renewable cost benchmarking reports globally |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Solar PV cost collapse**[149] | Utility-scale solar LCOE fell to $38–78/MWh ($20–45/MWh with PTC), making solar the cheapest source of new electricity in most of the world for 10+ years running per Lazard's annual benchmark. | Lazard LCOE+ · BloombergNEF Solar Module Index |
| **Lithium-ion battery price below $100/kWh**[148] | BloombergNEF's annual price survey logged $108/kWh for packs and $99/kWh for BEV packs in 2024 — the threshold at which a new EV reaches sticker parity with a comparable ICE vehicle without subsidies. | BNEF battery price survey |
| **LFP chemistry takeover**[146] | Lithium iron phosphate (LFP) has displaced NMC for most stationary storage and a growing share of EVs — cheaper, no cobalt, longer cycle life, the dominant chemistry in Tesla Megapacks and BYD vehicles. | CATL LFP · BYD Blade Battery · Tesla Megapack (LFP) |
| **Sodium-ion mass production**[145] | CATL Naxtra (~$40/kWh cell, 2026 mass production) and BYD's 30 GWh sodium-ion line are commercializing a chemistry that needs no lithium, no cobalt, no nickel — important for grid storage and emerging-market EVs. | CATL Naxtra · BYD sodium-ion · HiNa Battery |
| **Long-duration storage (iron-air)**[150] | Form Energy's iron-air batteries (~$250/kWh, 100-hour duration) target the multi-day grid-storage gap that lithium-ion is too expensive to fill — first utility deployments with Xcel and Georgia Power are live. | Form Energy iron-air · EnerVenue metal-hydrogen · ESS Inc iron flow |

## b) Current research frontier (last 12–18 months)

**01.** BloombergNEF Battery Price Collapse (June 2025): Lithium-ion pack prices fell 8% to record low of $108/kWh; stationary storage at $70/kWh (45% drop YoY); BEV packs at $99/kWh (2nd year below $100/kWh threshold)[148]

**02.** Solar & Wind LCOE Competitiveness (Lazard 2025): Utility-scale solar $38-78/MWh; onshore wind $37-86/MWh; with PTCs solar drops to $20-45/MWh; wind and solar least-expensive generation 10th consecutive year[149]

**03.** CATL Sodium-Ion Battery (April 2026): Naxtra technology at $40/kWh cell level (20% cheaper than LFP); mass production targeting 2026; operating range -40°C to 70°C; 10,000 cycle life with 90% retention at -40°C[145]

**04.** Form Energy Iron-Air Battery (2024-2025): Commercial-scale factory operational in West Virginia; utility deals with Xcel Energy, Southern Company, Georgia Power; target ~$250/kWh energy (reservoir); 100-hour duration; first deployments Q4 2025[150]

**05.** Tesla Megapack Deployment Surge (Q1 2026): Record 10.4 GWh deployed in Q1 2026 (156% YoY growth); full-year 2025 deployments 31.4 GWh (2.1x 2023); Shanghai Megafactory producing 1 GWh+ annually by Q2 2025[151]

## c) Who could be next

**BYD Sodium-Ion** [STARTUP][146] — BYD (battery and EV manufacturer) · Internal BYD funding; $37B+ market cap; profitable EV/battery leader

30 GWh sodium-ion factory breakground (Jan 2024); Q3 2025 delivery timeline; targeting <70% of LFP cost long-term; competing with CATL for sodium-ion market leadership

**Northvolt (now Lyten acquisition)** [STARTUP][152] — Northvolt (Swedish battery manufacturer) / Lyten (US battery company acquirer) · Lyten-backed restructuring; previously $5B+ valuation

Bankruptcy restructuring (Nov 2024, Mar 2025); Lyten acquisition of Northvolt assets including 16 GWh capacity; Northvolt Ett resuming production Q4 2025; European manufacturing recovery watch

**Envision AESC / HiNa Battery Technology** [STARTUP][153] — Envision AESC (joint venture); HiNa Battery Tech (Chinese sodium-ion leader) · Corporate backing (Envision Group, HiNa venture funding)

Sodium-ion alternatives gaining traction; HiNa targeting mass production 2025-2026; diversifying battery chemistry supply chains away from LFP dominance

## Flags / unverified

■ Northvolt status in flux (bankruptcy restructuring as of May 2026); actual Form Energy commercial deployment timing contingent on Q4 2025 utility project completion. All other figures from BloombergNEF, Lazard, Tesla official reports, and industry tracking verified.

■ Updated Firm 24/7 solar+storage LCOE (best regions): new value = $54-82/MWh

# Self-driving cars / robotaxis

**Impact so far.** Robotaxis crossed from pilot to scaled commercial reality between 2023–2025 — Waymo passing 4M paid driverless trips with peer-reviewed safety data, Tesla launching FSD-based robotaxis, and Aurora running fully driverless freight — finally validating a 15-year capital cycle.

**Watch for next.** Whether end-to-end neural-net autonomy (Tesla, Wayve) catches up with Waymo's HD-map / LiDAR stack on safety; expansion of Waymo One beyond 5–10 cities; Aurora's commercial trucking ramp; and whether Chinese AV scale (Apollo Go, Pony) fragments the global market.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Tekedra Mawakana**[154] | Co-CEO<br>Waymo (Alphabet) | Biz | — |
| **Dmitri Dolgov**[154] | Co-CEO and CTO<br>Waymo (Alphabet) | Biz | — |
| **Ashok Elluswamy**[155] | Director of Autopilot / FSD<br>Tesla | Biz | — |
| **Sebastian Thrun**[156] | Founder; Stanford Professor (emeritus from CS faculty)<br>Kitty Hawk / Stanford University; founded Google self-driving project (2009) | Cross | ★ DARPA Grand Challenge winner 2005 |
| **Chris Urmson**[157] | Co-founder and CEO<br>Aurora Innovation; previously CTO of Google Self-Driving Car (now Waymo) | Cross | — |
| **Raquel Urtasun**[158] | Founder and CEO; Professor<br>Waabi; University of Toronto | Cross | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Waymo's mapped + LiDAR robotaxi stack**[159] | HD maps + multi-sensor (LiDAR/radar/camera) + rules-and-learning hybrid is the only stack with peer-reviewed safety data at commercial scale and with paid driverless rides in 5+ US cities. | Waymo One (Phoenix, SF, LA, Austin, Atlanta) · Waymo Driver |
| **End-to-end neural-net driving**[160] | Replace hand-coded planners with a single learned model trained on video — Tesla FSD v12+, Wayve LINGO/GAIA, Comma.ai. Same scaling-laws logic as LLMs, applied to driving. | Tesla FSD v12+ · Wayve LINGO-2 / GAIA-2 · Comma.ai openpilot |
| **Autonomous trucking**[161] | Long-haul freight on highways is structurally easier than urban robotaxis; Aurora launched commercial driverless freight (Dallas–Houston) in May 2025, with Kodiak and Waabi following — a near-term economic prize. | Aurora Driver · Kodiak Robotics · Waabi Driver |
| **Generative-AI simulation**[162] | Neural-rendering simulators (Waabi World, NVIDIA Cosmos, Wayve GAIA-1) generate synthetic edge cases at scale — closing the rare-event data gap that pure on-road miles can't fill. | Waabi World · NVIDIA Cosmos · Wayve GAIA-1 · Applied Intuition |
| **Chinese robotaxi at scale**[163] | Baidu Apollo Go, Pony.ai, and WeRide run commercial driverless services across multiple Chinese cities; Apollo Go has reported 9M+ cumulative rides — a parallel deployment story to the US Waymo arc. | Baidu Apollo Go · Pony.ai · WeRide |

## b) Current research frontier (last 12–18 months)

**01.** Waymo One scaled paid driverless rides from a single Phoenix geofence (2020) to commercial operation across Phoenix, San Francisco, Los Angeles, Austin, and Atlanta, surpassing 4 million paid driverless trips by early 2025 and on track for 10M+ in 2026.[159]

**02.** Tesla launched its initial robotaxi service in Austin in June 2025 using FSD-only / vision-only AI inference (no LiDAR / HD maps), the first major commercial alternative to Waymo's stack.[164]

**03.** Waymo's safety data (Aug 2024 paper, 7M+ rider-only miles) reported 81% fewer airbag-deploying crashes and 78% fewer injury-causing crashes vs. a human-driven benchmark, the first peer-reviewed at-scale evidence that robotaxis can be safer than humans.[165]

**04.** End-to-end neural-net driving (Tesla FSD v12+, Wayve, Comma.ai) replaced hand-coded planners with a single learned model trained on video, mirroring the LLM-style scaling-laws approach in the autonomy stack.[160]

**05.** Chinese robotaxi operators (Baidu Apollo Go, Pony.ai, WeRide) are running commercial driverless services across Beijing, Wuhan, and Guangzhou; Apollo Go reported 9M+ cumulative rides by mid-2025.[163]

## c) Who could be next

**Wayve** [STARTUP][166] — Wayve AI Ltd. · Series C, $1.05B led by SoftBank (May 2024); also NVIDIA, Microsoft

Pioneering end-to-end embodied-AI autonomy; raised $1.05B Series C in May 2024 led by SoftBank, with NVIDIA and Microsoft participating; partnerships with Uber and OEMs to ship as a software-only AV stack.

**Waabi** [STARTUP][167] — Waabi Innovation Inc. · Series B, $200M led by Uber & Khosla (June 2024)

Raquel Urtasun's startup is building generative-AI simulation-first autonomous trucking; raised $200M Series B (June 2024) led by Uber and Khosla, targeting fully driverless trucking deployment in 2026.

**Aurora Innovation** [STARTUP][161] — Aurora Innovation, Inc. (NASDAQ: AUR) · Public (NASDAQ: AUR); ~$2B in cash reserves

Public company; launched the first fully driverless freight runs between Dallas and Houston in May 2025 — a leading indicator that long-haul autonomous trucking ships before passenger autonomy generalizes.

**Applied Intuition** [STARTUP][168] — Applied Intuition Inc. · Series F-equivalent, $250M at $15B valuation (2025)

Simulation and ADAS infrastructure provider used by most Western OEMs; raised $250M at $15B valuation (2025), positioning as the AWS of autonomy software.

## Flags / unverified

■ Tesla robotaxi rollout used safety operators in Austin in 2025 — fully driverless status varies by report

■ Waymo cumulative ride numbers cited from company-published statistics; independent audit limited

# Humanoid robotics

**Impact so far.** Humanoid robotics moved from research demos to low-volume industrial production between late 2024 and May 2026 — Tesla and Boston Dynamics in factories, Figure on the BMW line, 1X taking consumer pre-orders, with VLA models (RT-2, $\pi$0, Helix) finally enabling task generalization beyond hand-coded behaviors.

**Watch for next.** Whether Helix-class full-body autonomy generalizes to unstructured homes (1X NEO, Figure 03), whether teleop data factories drop unit-economic costs below $50/hour, and whether Tesla Optimus V3 actually hits 1M units/year — the inflection from industrial pilot to mainstream tooling.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Sergey Levine**[169] | Co-founder & Research Lead<br>Physical Intelligence | Biz | — |
| **Jitendra Malik**[170] | Professor, Robotics Researcher<br>UC Berkeley EECS | Acad. | — |
| **Dieter Fox**[171] | Senior Director, Robotics Research<br>NVIDIA / Allen Institute for AI | Cross | — |
| **Pieter Abbeel**[172] | Robotics & RL researcher<br>Amazon / UC Berkeley | Cross | — |
| **Brett Adcock**[173] | CEO & Founder<br>Figure AI | Biz | — |
| **Marc Raibert**[174] | Founder<br>Boston Dynamics; Boston Dynamics AI Institute | Biz | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Vision-Language-Action (VLA) foundation models**[175] | End-to-end neural policies that combine VLM backbones with robot action tokens — RT-2, RT-X / Open-X-Embodiment, Physical Intelligence $\pi$0/$\pi$0.5, and Figure's Helix enable zero-shot generalization across embodiments and tasks. | Google DeepMind RT-2 · RT-X / Open X-Embodiment · Physical Intelligence $\pi$0 / $\pi$0.5 · Figure Helix 02 |
| **Tactile sensing & dexterous manipulation**[176] | Multi-axis fingertip and palm sensors (3–5 mN sensitivity) fused with vision close the in-hand-manipulation gap — slip detection, contact-rich assembly, and delicate grasping become tractable. | Sanctuary AI Phoenix 75-DOF hands · Figure 03 fingertip + palm sensors · Toyota Punyo soft-skin · PaXini DexH13 |
| **Sim-to-real RL & whole-body control**[177] | Domain-randomized RL trained in Isaac Lab/Sim plus hybrid imitation pipelines now transfer zero-shot to hardware, with whole-body controllers handling balance and locomotion across embodiments. | NVIDIA Isaac GR00T N1.6 · Boston Dynamics electric Atlas · Humanoid-Gym · Fourier GR-1 |
| **Data factories & teleoperation scaling**[178] | Large-scale teleop pools (45–50 tutors per 100 robots) and at-home gig-economy recording have driven cost-per-tutor-hour down ~60% to ~$118 in 2026, accelerating data-hungry VLA training. | Tutor Intelligence DF1 · Hyundai Robotics Metaplant · Scale AI / Encord teleop · DoorDash chore data |
| **Form-factor convergence (26–32 DOF bipedal)**[179] | The industry has converged on ~26–32 DOF bipedal designs with 16–22 DOF tendon-driven hands, enabling vertical integration and shipment scaling toward 90K+ units in 2026. | Tesla Optimus V3 · Boston Dynamics electric Atlas · Figure 03 · Unitree G1 |
| **MolmoAct2 (open VLA)**[180] | Open-weight Vision-Language-Action foundation model for bimanual humanoid manipulation, trained on 720 h of teleoperation, with public weights, dataset, and eval code. | 87.1% real-world DROID success with unseen objects · ~37x faster inference vs MolmoAct1 · Open weights + 720 h dataset released |

## b) Current research frontier (last 12–18 months)

**01.** Figure's Helix 02 (Jan 2026) shows unified full-body autonomy from a single VLA — dishwasher loading, bottle uncapping, pill extraction, syringe dispensing — without task-specific training.[181]

**02.** Tesla Optimus V3 entered internal factory deployment in Jan 2026 with ~1,000 robots in production at Fremont and Gigafactory Texas; Tesla targets mass production at 1M units/year by Q3 2026.[182]

**03.** 1X NEO took 10,000 pre-orders in 5 days (Oct 2025); the Hayward, CA factory began full-scale production in Q1 2026 with consumer delivery targeted for late 2026.[183]

**04.** Unitree G1 launched at $16K (April 2026) with 23 DOF; Unitree filed for a Shanghai IPO at ~$610M (March 2026), aiming for 20K shipments in 2026 (vs ~5.5K in 2025).[184]

**05.** Boston Dynamics' electric Atlas entered commercial production in Jan 2026 with all 2026 units committed to Hyundai and Google DeepMind, supported by autonomous battery-swapping for continuous operation.[185]

## c) Who could be next

**Figure AI** [STARTUP][173] — Figure AI · Series C ~$1.5B at $39.5B valuation (Sep 2025); ~$1.9B total raised

Now fully independent from OpenAI on its own Helix VLA stack; reported $39.5B valuation (Sep 2025) and BMW production deal — a likely IPO candidate.

**Unitree Robotics** [STARTUP][184] — Unitree Robotics · Series C ~$520M at $5.5B (Feb 2026); IPO ~$610M filing

Best price/performance leader globally (G1 at $16K); Shanghai IPO filing March 2026 unlocks ~$610M for 4× capacity expansion targeting 20K units in 2026.

**Physical Intelligence** [STARTUP][186] — Physical Intelligence (Pi) · Founded 2024 by Levine, Abbeel et al; venture-backed (specifics undisclosed)

The leading VLA foundation-model lab; π0.5 (April 2026) shows emergent open-world generalization, positioned to license to OEMs across the industry.

**Sanctuary AI** [STARTUP][187] — Sanctuary AI · Series B ~$50–100M (estimate); Magna strategic partnership

75-DOF hydraulic hands with industry-leading dexterity and a Magna automotive partnership for production deployment.

**Figure 03 production scale-up** [STARTUP][188] — Figure AI · Series C+; ~$39B reported valuation

First humanoid OEM to credibly hit hourly production cadence with autonomous (non-teleop) locomotion on uneven terrain - a leading indicator of commercial humanoid roll-out.

## Flags / unverified

■ Tesla Optimus V3 specs and production timeline come largely from non-Tesla sources; treat with caution.

■ Figure's claims about full Helix capabilities are demo-tested, not peer-reviewed for unseen-task generalization.

■ Humanoid market projections (90K shipments 2026, 1.2M by 2030) are highly speculative.

# Fusion energy

**Impact so far.** Fusion went from public-science promise to active commercial track between 2022 and 2026: NIF demonstrated repeatable net energy gain, CFS broke ground on SPARC with HTS magnets, and Helion shipped a D-T plasma machine on the way to a 50 MW commercial PPA with Microsoft.

**Watch for next.** First private-fusion net-electricity-on-grid demonstrations (CFS SPARC Q>1 in 2027, Helion Orion 2028), AI plasma control becoming standard, and validation of tritium breeding fuel-cycle infrastructure at commercial scale.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Omar Hurricane**[189] | Chief Scientist, Inertial Confinement Fusion Program<br>Lawrence Livermore National Laboratory | Gov | — |
| **Bob Mumgaard**[190] | Co-founder & CEO<br>Commonwealth Fusion Systems | Biz | — |
| **David Kirtley**[191] | Co-founder & CEO<br>Helion Energy | Biz | — |
| **Michel Laberge**[192] | Founder & CTO<br>General Fusion | Biz | — |
| **Pietro Barabaschi**[193] | Director-General<br>ITER Organization | Gov | — |
| **Sam Altman**[194] | Chairman of the Board<br>Helion Energy (also CEO, OpenAI) | Cross | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Tokamak (magnetic confinement)**[195] | The most-developed fusion approach: a toroidal magnetic chamber confines a hot plasma. CFS SPARC, ITER, KSTAR/EAST, JT-60SA, and DIII-D are the major active programs. | CFS SPARC (HTS magnets) · ITER · KSTAR (100M°C, 48s record) · EAST (403s H-mode) |
| **Inertial confinement (laser-driven)**[196] | High-energy lasers compress a fusion fuel pellet to ignition. NIF reached repeatable net energy gain (8.6 MJ from 2.08 MJ in April 2025) — the first machine of any kind to do so. | NIF (LLNL) · Marvel Fusion (proton-boron) · Pacific Fusion |
| **HTS (high-temperature superconductor) magnets**[197] | Rare-earth barium copper oxide (REBCO) tape enables 20+ tesla magnetic fields, cutting tokamak size and cost dramatically — the enabling technology for compact commercial fusion machines. | CFS 20-tesla TF magnet · Tokamak Energy ST40 / Demo4 · MIT-CFS HTS collaboration |
| **AI plasma control**[198] | Deep RL controllers stabilize plasma in real time on DIII-D, KSTAR, and TCV; DeepMind's TORAX simulator (May 2024) is now part of CFS's planning loop and broadly adopted across the field. | DeepMind TORAX · DeepMind + EPFL TCV (Nature 2024) · DIII-D + KSTAR ELM-suppression AI |
| **Alternative confinement concepts**[194] | Field-reversed configuration (Helion, 150M°C D-T, Feb 2026), sheared-flow Z-pinch (Zap Energy, 1.6 GPa pressures), stellarators (Wendelstein 7-X, 43s record May 2025), and acoustically driven MTF (General Fusion). | Helion Polaris (FRC) · Zap Energy FuZE-3 (Z-pinch) · Wendelstein 7-X (stellarator) · General Fusion LM26 (MTF) |

## b) Current research frontier (last 12–18 months)

**01.** NIF achieved 10+ ignition shots with a peak yield of 8.6 MJ (April 2025) — establishing repeatable net energy gain and a target-gain trend above 4:1 for inertial confinement fusion.[199]

**02.** Commonwealth Fusion Systems installed the first of 18 toroidal-field HTS magnets on SPARC in Jan 2026; first plasma is targeted for 2026 with a Q>1 demonstration in 2027.[200]

**03.** Helion Polaris reached 150 million °C with deuterium-tritium fuel (Feb 2026) — the first privately-built D-T fusion machine; Orion commercial plant construction began Jul 2025 with a 50 MW Microsoft PPA targeting 2028.[194]

**04.** KSTAR set a 100 million °C, 48-second plasma record (2024) with a 300-second target by 2026; Wendelstein 7-X reached a 43-second high-fusion-product plasma in May 2025.[201]

**05.** AI/RL plasma control was demonstrated on DIII-D and KSTAR in 2024; DeepMind's TORAX simulator (May 2024) is now embedded in CFS's daily operational planning and adopted across the field.[198]

## c) Who could be next

**Commonwealth Fusion Systems (SPARC)** [STARTUP][202] — CFS (MIT spinout) · $3.0B total; Series B2 $863M (Aug 2025) led by Nvidia, Google, Bill Gates

Largest private fusion effort: $3B+ raised, magnet assembly underway, first plasma targeted 2026 and Q>1 in 2027 — closest to a credible commercial tokamak.

**Helion Energy (Polaris / Orion)** [STARTUP][203] — Helion Energy · Series E+ via Sam Altman, Khosla Ventures; Microsoft PPA May 2023

Only private firm with a firm grid PPA (Microsoft 50 MW by 2028); FRC achieved 150M°C D-T (2026); Orion commercial plant under construction.

**TAE Technologies (Norm / Copernicus)** [STARTUP][204] — TAE Technologies · $1.3B+ equity raised; $150M latest round (Chevron, Google, NEA)

Neutral-beam injection breakthrough (Apr 2025) cut machine complexity ~50%; Copernicus targeting net energy before 2030; planned TMTG-style merger at $6B valuation (mid-2026).

**Tokamak Energy (ST40 / Demo4)** [STARTUP][197] — Tokamak Energy · Multiple rounds; UK government and private capital

Spherical tokamak architecture with 11.8-tesla HTS magnets; claims 30–40× smaller plasma volume vs LTS designs and a clear path to ST-E1 commercial pilot.

## Flags / unverified

■ Sam Altman is Chairman of Helion's board, not founder (David Kirtley founded the company).

■ TAE merger details and valuation are reports of plans, not finalized transactions.

■ Updated zap_energy_strategy: new value = Hybrid fission-fusion company strategy; Zabrina Johal named CEO (May 5)

# Brain-computer interfaces

**Impact so far.** BCIs crossed from lab to early commercial reality between 2024 and 2026: Neuralink put 5+ patients on the Telepathy implant, Precision Neuroscience took home the first wireless-cortical-BCI 510(k), BrainGate hit 110 cpm bimanual typing, UCSF restored speech at 78 wpm, and Meta shipped a 20.9 wpm sEMG wristband to consumers.

**Watch for next.** The first BCI Pre-Market Approval (likely Synchron), Neuralink's first Blindsight implant, Paradromics speech-restoration pivotal results, and whether the non-invasive consumer category (Meta Neural Band, Kernel) closes meaningful ground on invasive bandwidth.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Leigh Hochberg**[205] | Director, Center for Neurotechnology and Neurorecovery; IDE sponsor-investigator, BrainGate<br>Brown University / Massachusetts General Hospital | Cross | ★ Sean M. Healey International Prize for Innovation in ALS (2024) |
| **Edward Chang**[206] | Director, Speech Neuroprosthesis Lab<br>UCSF Department of Neurosurgery | Acad. | — |
| **Jaimie Henderson**[207] | Director, Stanford Neural Prosthetics Translational Lab; Co-PI, BrainGate2<br>Stanford School of Medicine | Acad. | — |
| **Thomas Oxley**[208] | Co-founder & CEO<br>Synchron Inc. | Biz | — |
| **Matt Angle**[209] | Founder & CEO<br>Paradromics | Biz | — |
| **Benjamin Rapoport**[210] | Co-founder & Chief Science Officer<br>Precision Neuroscience | Biz | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **Intracortical microelectrode arrays (Utah arrays)**[211] | High-density silicon arrays implanted directly into motor cortex (Blackrock NeuroPort) record from individual neurons — the gold standard for invasive BCIs since the early 2000s. | BrainGate clinical consortium · Blackrock NeuroPort · Stanford / Brown / MGH BrainGate2 |
| **Endovascular BCI (Stentrode)**[208] | Synchron's stent-electrode is implanted via blood vessels — no open-brain surgery — trading some signal fidelity for dramatically lower surgical risk. | Synchron Stentrode · FDA COMMAND trial · Synchron + Apple Vision Pro integration |
| **High-bandwidth thin-film cortical interfaces**[210] | Flexible subdural electrode films (e.g. Precision Layer 7) put 1,024+ electrodes on a hair-thin sheet, with FDA 510(k) clearance and a Medtronic surgical-navigation partnership in 2025–26. | Precision Neuroscience Layer 7 · Mount Sinai 4,096-electrode demo · Medtronic + Precision partnership |
| **Speech decoding & synthesis**[212] | UCSF's Chang lab and Stanford have pushed motor-cortex speech decoders to 78 wpm with personalized voice synthesis; the BrainGate consortium hit 110 cpm bimanual typing. | UCSF Chang lab speech BCI · Stanford inner-speech decoder · BrainGate bimanual typing |
| **Non-invasive consumer BCI (EMG / EEG / fNIRS)**[213] | Wearable surface-EMG and EEG/fNIRS systems (Meta Neural Band, Kernel Flow, OpenBCI) are reaching consumer scale at meaningful — if lower — bandwidths than invasive devices. | Meta Neural Band sEMG wristband · Kernel Flow TD-fNIRS · OpenBCI |

## b) Current research frontier (last 12–18 months)

**01.** Neuralink's PRIME trial has implanted 5+ patients across multiple sites by mid-2025; the second patient ('Alex') achieved CAD control and gaming within days, and Blindsight (visual prosthesis) received FDA breakthrough device designation (Sep 2024) with first-in-human implant planned for 2026.[214]

**02.** Paradromics received FDA IDE approval (Nov 2025) for the Connect-One speech-restoration trial with the Connexus BCI, claiming ~200+ bps information transfer (10–20× higher than BrainGate, 100–200× higher than Synchron) via thread arrays thinner than a human hair.[215]

**03.** INBRAIN Neuroelectronics completed enrollment for the world's first graphene-BCI clinical trial (April 2026) with zero device-related adverse events in the first cohort; 1,024-contact wireless rechargeable arrays.[216]

**04.** Precision Neuroscience's Layer 7 received FDA 510(k) clearance (April 2025) — the first wireless cortical BCI cleared for market — and is now in 30-day chronic trials at six U.S. medical centers; Medtronic partnership announced January 2026.[210]

**05.** BrainGate reported rapid bimanual typing (110 cpm, 1.6% WER) in two paralyzed patients (Nature Neuroscience, March 2026); the consortium raised $200M Series D in late 2025 for FDA PMA pivotal trials.[217]

**06.** Neuralink developing surgical robot for precision brain access with automated procedures and high-volume implant production targeted for 2026.[218]

**07.** Synchron's Stentrode is the first BCI integrated with Apple's BCI HID profile as a native input category, enabling hands-free Apple-device control for paralysis patients.[219]

## c) Who could be next

**Neuralink Blindsight** [STARTUP][220] — Neuralink · Reported $1B+ private equity, Musk-led

First visual-prosthesis implant planned in 2026 — direct visual-cortex stimulation that bypasses damaged optics. A major expansion from motor-only BCIs.

**Synchron pivotal trial** [STARTUP][208] — Synchron Inc. · $200M raise (Nov 2025) for pivotal trial execution

Preparing FDA Pre-Market Approval submission across 4+ U.S. sites; if approved, the first BCI PMA could define the regulatory gold standard for the field.

**Paradromics Connexus** [STARTUP][209] — Paradromics Inc. · Series-stage venture funding (specifics undisclosed)

Extreme bandwidth (~200+ bps), first FDA IDE for speech restoration as a primary endpoint, 3 clinical sites planned for early 2026.

**INBRAIN Neuroelectronics** [STARTUP][221] — INBRAIN (Barcelona) · Spanish/EU biotech funding; WEF recognition signals next round

Graphene-electrode material breakthrough with bi-directional 1,024-contact arrays; WEF Technology Pioneer 2025 and zero adverse events in first cohort.

## Flags / unverified

■ Paradromics' 200+ bps figure is a manufacturer claim; not yet independently verified in human use.

■ BCI regulatory pathway is fragmented across 510(k) / IDE / PMA tracks; no unified FDA BCI guidance exists yet.

# AI accelerators / advanced compute

**Impact so far.** GPU-dominated AI compute fractured into specialized lanes between 2024 and 2026 — wafer-scale (Cerebras), inference ASICs (Groq, Etched), photonic interconnect (Lightmatter), and per-hyperscaler custom silicon (TPU/Trainium/Maia/MTIA) — pulling 20–70% of opex out of training and inference.

**Watch for next.** Rubin (2H 2026) and the Cerebras OpenAI 750 MW deployment going live; Lightmatter Passage co-packaged optics in production; neuromorphic chips (Loihi 3, NorthPole) crossing the practical edge-AI threshold; whether NVIDIA-Groq integration cements inference dominance.

## a) People behind it

| Name | Role / Affiliation | Type | Prizes |
|---|---|---|---|
| **Jensen Huang**[222] | CEO & Co-founder<br>NVIDIA | Biz | — |
| **Andrew Feldman**[223] | Co-founder & CEO<br>Cerebras Systems | Biz | — |
| **Jonathan Ross**[224] | Founder<br>Groq (acquired by NVIDIA Dec 2025) | Biz | — |
| **Jim Keller**[225] | CEO<br>Tenstorrent | Biz | — |
| **Norm Jouppi**[226] | Engineering Fellow; lead architect, Google TPU<br>Google | Biz | ★ ACM-IEEE Eckert-Mauchly Award |
| **Nick Harris**[227] | Co-founder & CEO<br>Lightmatter | Biz | — |

## Sub-ideas inside this breakthrough

| Sub-idea | Description | Examples |
|---|---|---|
| **GPU scaling (Hopper → Blackwell → Rubin)**[228] | NVIDIA's flagship line keeps doubling effective AI throughput per generation: Blackwell (208B transistors, ~20 PFLOPS FP4 per B200) ships at scale in 2025, Rubin (3× Blackwell, 1.2 EFLOPS FP8 training) in 2H2026. | NVIDIA H100/H200 (Hopper) · NVIDIA B200 (Blackwell) · NVIDIA Rubin / Rubin Ultra |
| **Wafer-scale processors**[229] | Cerebras WSE-3 (5nm, 4 trillion transistors, 900K cores, 44 GB on-chip SRAM, 21 PB/s memory) eliminates chip-to-chip movement entirely — enabling single-chip training of multi-trillion-parameter models. | Cerebras WSE-3 · Cerebras CS-3 system · OpenAI 750 MW Cerebras deal |
| **Inference-specialized ASICs**[224] | Fixed-function silicon optimized for narrow inference workloads — Groq LPU's deterministic conveyor-belt architecture and Etched's transformer-only Sohu ASIC compete with GPUs on $/token. | Groq LPU (acquired by NVIDIA) · Etched Sohu · MatX |
| **Hyperscaler custom silicon**[230] | Cloud platforms have abandoned GPU monoculture: Google Trillium / TPU v6, AWS Trainium 3, Meta MTIA 300/500, Microsoft Maia 200 each deliver hyperscaler-specific compute economics. | Google TPU v6 Trillium · AWS Trainium 3 · Meta MTIA 300/500 · Microsoft Maia 200 |
| **Photonic & neuromorphic compute**[231] | Lightmatter's photonic processor and IBM NorthPole / Intel Loihi 3 neuromorphic chips break with the GPU paradigm — using photons or spiking neurons for orders-of-magnitude better energy per inference for the right workloads. | Lightmatter Envise / Passage · IBM NorthPole · Intel Loihi 3 |

## b) Current research frontier (last 12–18 months)

**01.** NVIDIA Rubin GPUs ship in 2H 2026 with ~3× Blackwell performance and 1.2 EFLOPS FP8 training capability; Rubin Ultra follows in 2027.[222]

**02.** Cerebras Systems signed a >$10B agreement with OpenAI (Jan 2026) for ~750 MW of inference compute through 2028 and filed for IPO at ~$23B valuation in April 2026.[232]

**03.** NVIDIA acquired Groq for ~$20B (Dec 2025) and integrated the LPU into its 2026 GTC stack; Groq 3 LPU achieves ~1,500 tokens/sec on agentic AI inference.[233]

**04.** Hyperscaler custom silicon scaled across all four majors: Google Trillium (4× LLM training perf vs v5e, 2× HBM bandwidth), AWS Trainium 3 (2.52 PFLOPS MXFP8, 144 GB HBM3e), Meta MTIA generations 4× HBM bandwidth, Microsoft Maia 200 (10 PFLOPS FP4).[230]

**05.** Intel Loihi 3 (Jan 2026) delivers 8M neurons / 64B synapses at 4nm running at ~1.2W peak vs 300W+ for GPU equivalents — neuromorphic compute crossing the threshold for practical edge AI.[234]

**06.** Cerebras' $3.5B IPO at ~$26.6B valuation, anchored by a $20B+ OpenAI compute deal, validates wafer-scale silicon as a credible alternative to GPU dominance for AI inference.[235]

## c) Who could be next

**Tenstorrent** [STARTUP][236] — Tenstorrent · Series B (2024) led by Khosla Ventures and others

Jim Keller's open-source RISC-V + Metalium ASIC pursuing GPU alternatives for hyperscale clusters via 12×400 Gbps Ethernet scale-out; Blackhole successor scaling.

**Rebellions** [STARTUP][237] — Rebellions Inc. · Series C 2025, $1.4B valuation; Arm + Samsung Ventures + Synopsys

Korean accelerator with 4nm UCIe-Advanced quad-chiplet architecture (REBEL-Quad), 144 GB HBM3e, claiming ~3.2× tokens/W vs H200; Arm + Samsung backed.

**FuriosaAI (RNGD next-gen)** [STARTUP][238] — FuriosaAI · Rejected $800M Meta acquisition; raised capital via LG and OpenAI partnerships

Korean TSMC 5nm AI ASIC; founder declined Meta's $800M acquisition (March 2025) and is pursuing a 2027 IPO with LG and OpenAI partnerships.

**Lightmatter (photonic compute)** [STARTUP][231] — Lightmatter · Series D ~$400M (2024); SoftBank Vision Fund, GV

Shipping the Envise photonic processor and Passage L200/L20 co-packaged optics — the leading commercial bet on photonic interconnect for AI datacenters.

## Flags / unverified

■ Etched Sohu's 500K tokens/sec figure is a manufacturer claim and the part remained unshipped at the time of writing.

■ FuriosaAI declined-acquisition story is widely reported but specifics are second-hand.

# Five cross-cutting themes

**1. Scaling laws are the through-line.**

Every one of the AI breakthroughs (transformers, reasoning models, diffusion) explicitly relies on a scaling-law thesis. Battery and solar cost-down curves (Wright's law) are the same shape: roughly 20% cost reduction per doubling of cumulative volume. Reusable rocketry is the same curve in cost-per-kg-to-orbit.

**2. Verifiable rewards close the loop.**

Reasoning models train on tasks with checkable answers. AlphaFold is verified against crystal structures. QEC has a hard threshold. CRISPR and obesity Phase-3 trials hit pre-registered endpoints. Where the reward is dense and verifiable, RL — and capital — pour in.

**3. Delivery is the new bottleneck in biology.**

mRNA vaccines, in vivo CAR-T, in vivo CRISPR, and personalized cancer vaccines share one stack: a payload wrapped in a lipid nanoparticle. Whoever solves non-hepatocyte targeted delivery unlocks the rest of the genome.

**4. Reusability and amortization beat raw novelty.**

Falcon 9's moat is the same booster flying 22+ times. MoE LLMs reuse a small fraction of parameters per token. Surface-code logical qubits trade more physical qubits for arbitrarily long lifetime. Battery cycle life, not raw energy density, is what brought $/kWh-cycle below grid parity.

**5. The lab → spinout → IPO path is now standard.**

AlphaFold launched Isomorphic Labs. Baker's IPD seeded Xaira / Generate / Outpace. Karikó & Weissman seeded BioNTech. Doudna & Charpentier seeded CRISPR Tx, Editas, Intellia. Liu's lab spawned Beam, Prime, and Editas. Sutton & Barto's RL textbook seeded an entire research lineage now valued at hundreds of billions.

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